# Design of PLC Based Automatic Dam Shutter Control

**Vaibhav Pawar[1] Maya Potdar[2] Kajal Rajguru[3] Ajay Sargar[4] Pavan Bhujbale[5]**
[1,2,3,4,5]Department of Electrical Engineering
[1,2,3,4,5]Zeal College of Engineering & Research, Pune, India

*Abstract* — This paper aims to provide an automization of dam gates. This paper presents the design and implementation of a PLC (Programmable Logic Controller) based automatic dam shutter control system aimed at improving the efficiency and safety of dam operations. The system utilizes PLC technology to automate the opening and closing of dam shutters based on water level readings, thereby reducing the risk of human error and ensuring timely response to changing water levels. In the proposed system various parameters such as level and limit are controlled. In the proposed system the PLC plays the main role. PLC controls the devices and input switches and signals give a controlling signal to the PLC. In the proposed system, the PLC used is Allen Bradley SLC 500/03. This PLC has 16 digital I/O and 8 analog I/O.

***Keywords:*** PLC, Ladder logic, Float sensor, Automation

## I. INTRODUCTION

The water availability is very critical in every sector like, public, industry, power sector etc. The paper gives an idea about efficient monitoring and controlling of water level in dam. In traditional system the water level of dam is managed by manually which requires large man power. This makes it very costly and inaccurate. PLC serves various features like, flexibility, digital nature, versatility. So this features of PLC select to automate the dam gates. The PLC controls the dam gate automatically.

In Our India there are approximately 3200 dam are present. In Gujarat, 202 dams are there out of which 95 dams have gates. Approximately, these dams cover 1,70,000 sq.km catchment area for collecting water. There is also 2067.68 km long and complex canal network through which about 10 lakes hectares land gets water for irrigation and drinking purpose. Only 1 dam has partial automated dam controlling system (Ukai dam on Tapi River at Surat). In all over India only 1 canal has fully automated gates. (Indira Gandhi canal on HaraikiBeraj reservoir).

## II. BLOCK DIAGRAM

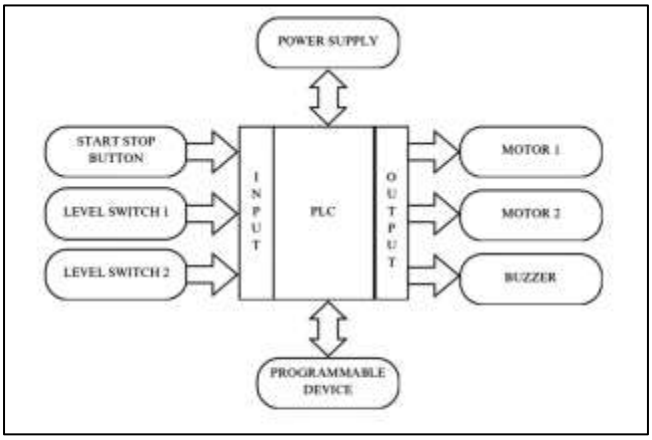



Fig. 1: Block Diagram of PLC Based Dam Shutter Control

## III. METHODOLOGY

The main components of proposed system includes PLC, DC motor and float sensor.

### A. *Programmable Logic Controller (PLC)*

This is the main controlling element of the proposed system. The PLC controls the operation of the system. This is modular SLC 500/03 PLC and offers several I/O configurations. In this PLC, there are 16 digital I/O and 8 analog I/O are available.

In a given system, two float sensors are the input switches connected to the input of the PLC which gives an input signal to the PLC. SMPS is connected to give PLC a 24V supply. The output of the PLC is connected to the DC motor. This motor is connected to the relay. The relay rotates the motor in two directions which we require for forward and reverse operation of dam gates. This all the system works by ladder programming.

Ladder programming is one of the programming languages of PLC. It is used in industrial control applications.

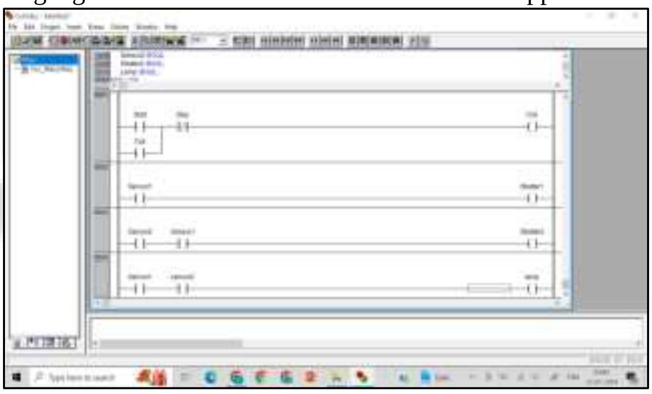


Fig. 2: The ladder programming of the system

Here in this ladder diagram, the conditions of gate closing and opening are given. When the supply has started the mains will turn on at the start of the program. After that during the starting condition, the water level is below the low-level switch then the green LED will blow. After that when water touches the low-level switch it gives input PLC and then motor 1 rotates in the forward direction and opens gate 1. When the water reaches a high-level switch it gives input to PLC and motor 2 rotates in the forward direction and opens gate 2.

When both level switches turn on buzzer will turn on. So after the buzzer turns on, both the motors will rotate in reverse direction and turn off both the gates.

### B. *Float sensor*

In the proposed system the float sensor is the sensing element. The float sensor gives the input signal to the PLC. Two float sensors are used which are at the lower and upper levels. When the water touches the float sensor it gives a signal to the PLC to open a dam gate. It controls the action of the PLC as open and closed.

A float sensor can be used to measure the water level in the dam. The float sensor consists of float that moves up and down with the water level changes. As the float moves, it triggers a switch or sensor element that sends a signal to the PLC indicating the current water level.

*1) Working of float sensor*

The float is typically on a vertical rod or cable that allows the float to move up and down with the water level changes. As the water level in the dam changes, the float moves accordingly. When the float reaches a certain level, it triggers a switch or sensor element that generates an electrical signal. This signal is then sent to the PLC input module for processing. The PLC program is designed to read the input signal from the float sensor and the PLC makes decisions based on water level. The PLC automatically opens or closes the dam gates to regulate the flow of water.

### *C. Gate control mechanism*

In gate control system there are two doors in our system. One gate is for prior level and other is used for highest level. When the water sensed by sensor 1 the gate 1 door will be opened by DC motor.

When the water sensed by the sensor 2 the gate door will be opened by another DC motor. The gate will open and close by DC motor which rotates in reverse and forward direction with help of relay.

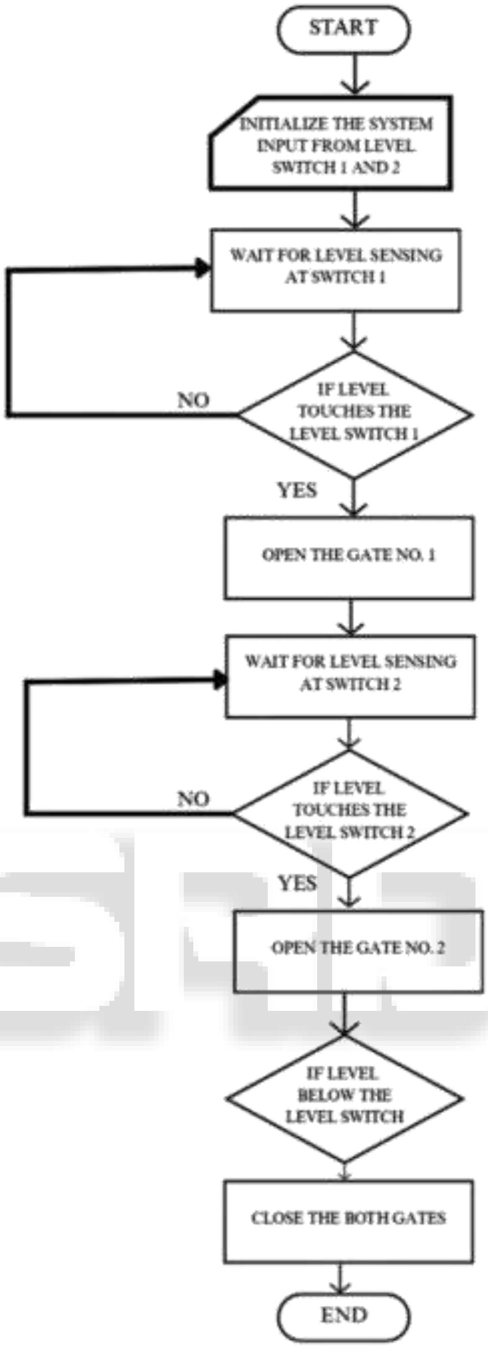
The gate control mechanism involves using the PLC to monitor and control the opening and closing of the dam gates. The PLC receives inputs from the float sensors. These sensors provide level of water at the dam. The PLC runs a control logic program that processes the sensor input and determine whether to open or close dam gates. The control logic program can be developed by ladder logic programming. The PLC sends control signal to the output of the PLC which are responsible for opening and closing of dam gates.

Safety interlocks are implemented in the control system to ensure the safe operation of the dam gates. These interlocks could be buzzer to prevent accident or damage to the equipment.

By integrating these components and functionalities, PLC based automatic dam shutter control system can effectively manage the operation of dam gates based on control algorithm, ensuring efficient water flow management and safety at the dam site.

## IV. FLOWCHART

In this flowchart there are three conditions are given. At the start wait for level sensing at switch 1 and when level is sensed by switch 1 it opens the gate 1 as PLC sends the command to the motor to rotate in forward direction. If level is not sensed by switch 1 wait for it. After that wait for level sensing at switch 2 and when level is sensed by switch 2 it opens the gate 2 as PLC sends the command to the motor to rotate in forward direction. If level is not sensed by level switch 2 wait for it. when water goes below level switch turns off the gate.




## V. FUTURE SCOPE

To enhance the sophistication of this process we can make use of a level transmitter and standalone PID controller. This system opens and closes within particular stages of gates.
The level transmitter can be used by RFID devices for wireless communication along with the PLC. In this system, we use also GPS. Global position systems indicate that a particular person will receive messages and alerts through mobile.

## VI. CONCLUSION

The proposed system offers a sophisticated and efficient solution for managing the operation of dam gates. The system demonstrates the benefits of automation in water management systems, offering improved efficiency, accuracy, and safety in controlling dam operations.

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